Machine Learning Lab- Laboratory Record

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Abstract

This project analyses the academic performance of 649 secondary school students using the Student Performance dataset (Portuguese subject). The goal is to identify key factors influencing students' final grades (G3) and build machine learning classification models to predict whether a student will pass or fail. Exploratory data analysis was performed across areas including study habits, parental education, alcohol consumption, social life, and health. Three classification models - Logistic Regression, Random Forest, and Gradient Boosting were trained on engineered features. The project demonstrates that prior grades (G1, G2), number of failures, and study time are among the strongest predictors of academic outcome.

Introduction

Problem Statement

Academic failure is a significant concern in educational systems worldwide. Early identification of students at risk of failing can enable timely intervention by teachers, parents, and school administrators. However, student performance is influenced by a complex mix of factors not just academic history, but also lifestyle choices, family background, and social behaviour. The challenge lies in understanding which of these factors matter most, and building a reliable predictive system.

Importance of the Topic

With increasing access to student data, machine learning offers a powerful tool to extract meaningful patterns and make data-driven decisions. Predicting student outcomes using features like study time, parental education, alcohol consumption, and prior grades can help educators personalise support and reduce dropout rates. This project addresses this problem using real-world data from two Portuguese secondary schools.

Objective

The primary objectives of this project are:

- To perform a thorough exploratory data analysis (EDA) on the student dataset
- To identify features most correlated with final grade (G3)
- To engineer meaningful new features from existing data
- To build and evaluate machine learning models that classify students as Pass ($G3 \geq 10$) or Fail ($G3 < 10$)

Dataset Overview

The dataset contains 649 student records with 33 original features, covering demographics (age, sex, address), family background (parental education, family size), school-related attributes (study time, failures, extra support), and lifestyle variables (alcohol consumption, going out, health, absences). The target variable for classification is whether a student passes ($G3 \geq 10$).

Methodology/ Working

Step 1: Data Loading and Inspection

The dataset (student-por.csv) was loaded using pandas. It contains 649 rows and 33 columns with no missing values across any feature, confirmed via df.isnull().sum(). Features include 16 integer columns and 17 object (categorical) columns.

Step 2: Exploratory Data Analysis (EDA)

Several categories of analysis were performed:

Grade Distribution: The distributions of G1, G2, and G3 were plotted. Average grades across the three terms were 11.40, 11.57, and 11.91 respectively, showing a slight upward trend. Grade ranges ran from 0 to 19.

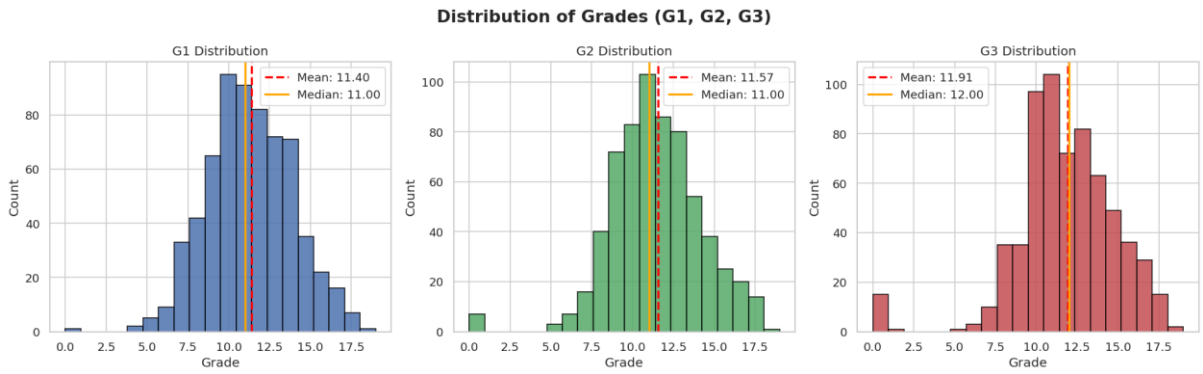

Fig 1. Distribution of Grades G1, G2, G3)

Categorical Feature Analysis: Average G3 was computed across 11 categorical features including sex, address, internet access, higher education aspiration, and romantic relationship status. Students aspiring to higher education had a notably higher average G3 (12.28) compared to those who did not (8.80).

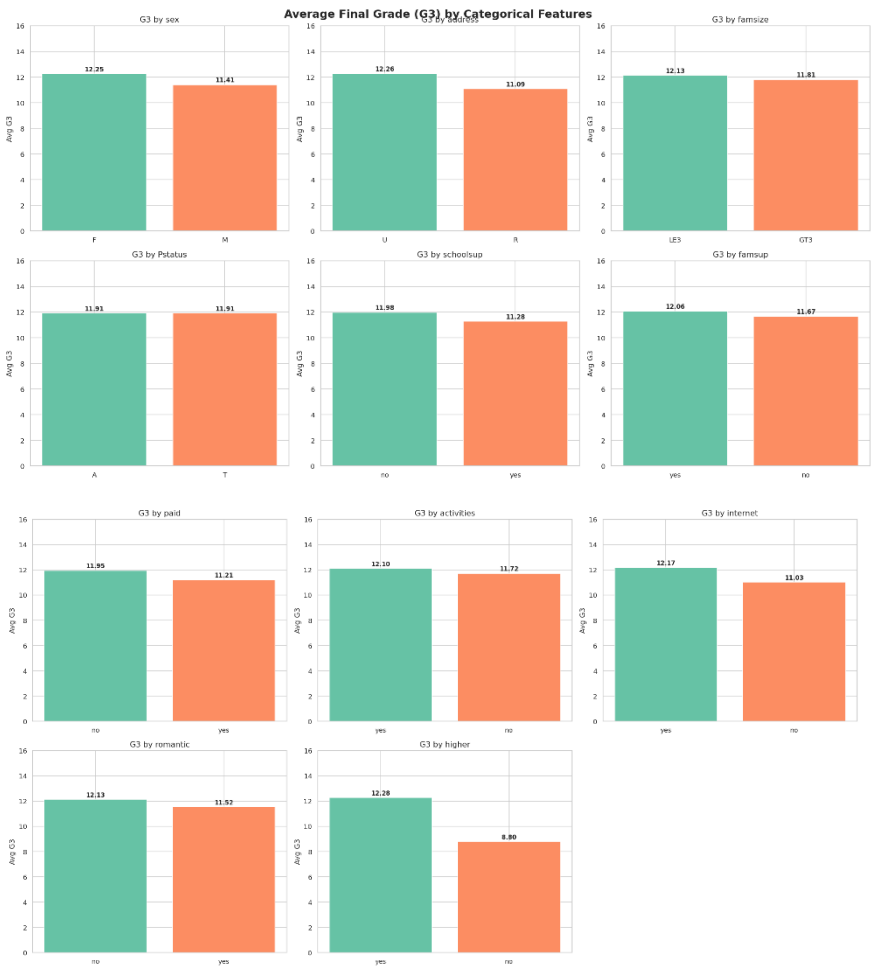

Fig 2. Average Final Grade by Categorical Features

Correlation Analysis: A heatmap of numerical features revealed that G2 (r = 0.92) and G1 (r = 0.83) had the strongest positive correlations with G3. Among negative correlators, failures (r = −0.39) and daily alcohol (Dalc, r = −0.20) were most significant.

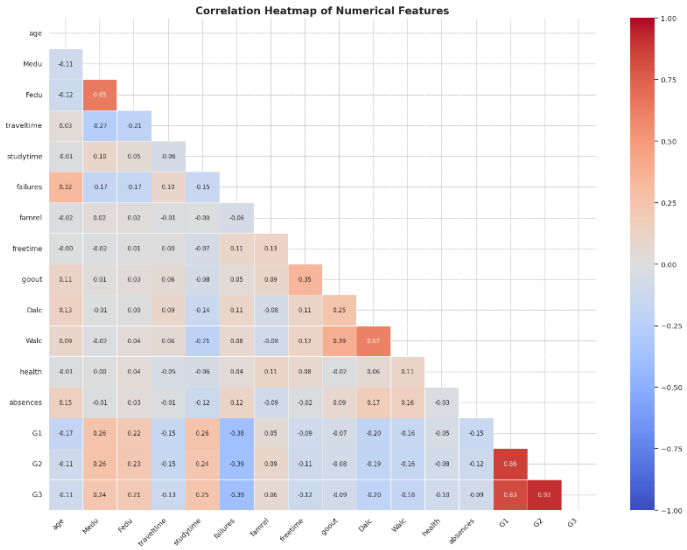

Fig 3. *Correlation Heatmap*

Alcohol Consumption Analysis: Both workday (Dalc) and weekend (Walc) alcohol consumption showed negative correlations with G3 (−0.205 and −0.177 respectively). Males consumed notably more alcohol on average than females.

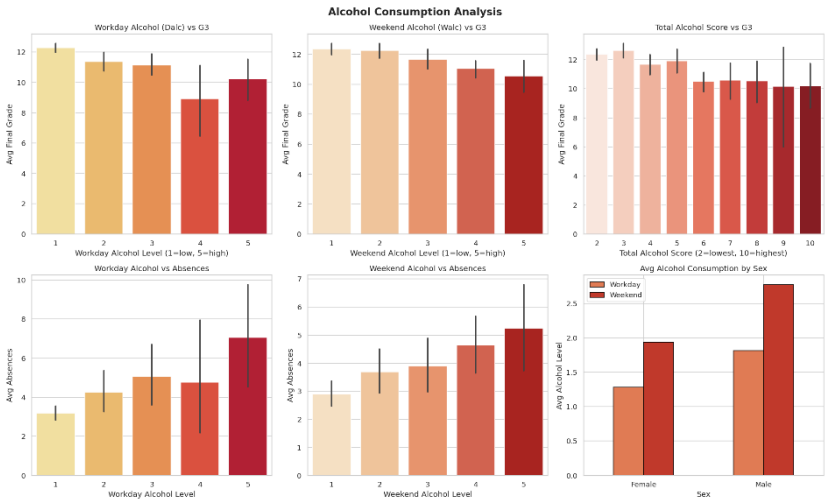

Fig 4. Alcohol Consumption Analysis

Parental Education Analysis: Students whose mothers had higher education levels scored better on average (up to 13.07 at the highest level). A combined parent education score showed a positive trend with G3.

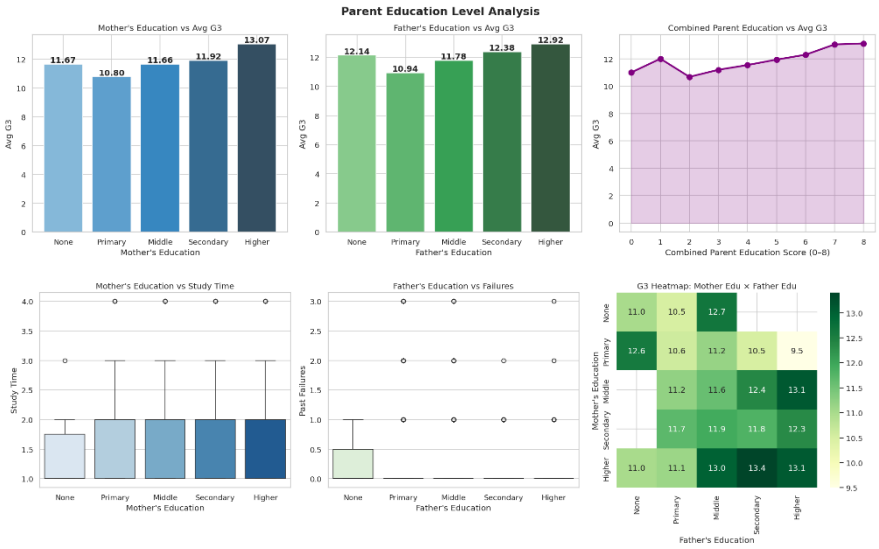

Fig 5. Parent Education Level Analysis

Failures and Study Time Analysis: Most students (549 out of 649) had zero prior failures. Students with 0 failures averaged around 12+ in G3, while those with 3 failures averaged as low as 7–8. Study time had a positive effect, with students studying over 5 hours averaging G3 scores of 13+.

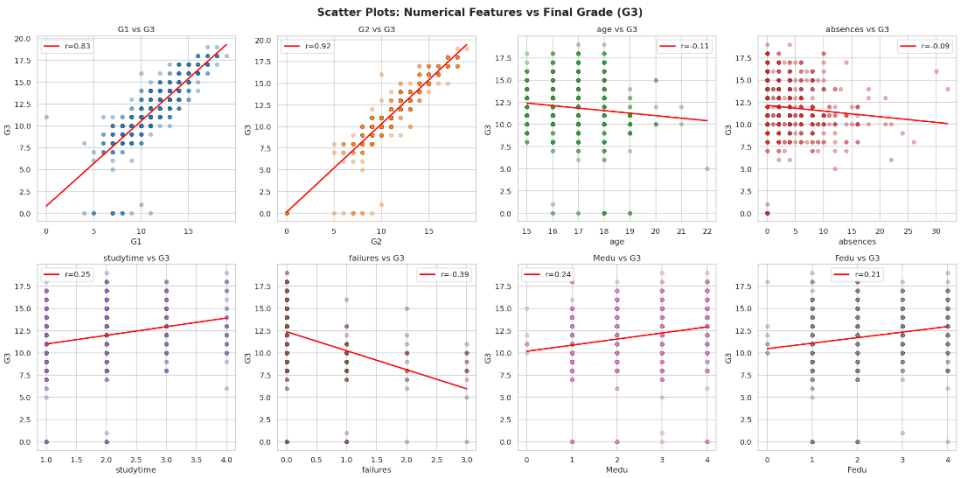

Fig 6. Scatter Plots: Numerical Features vs G3

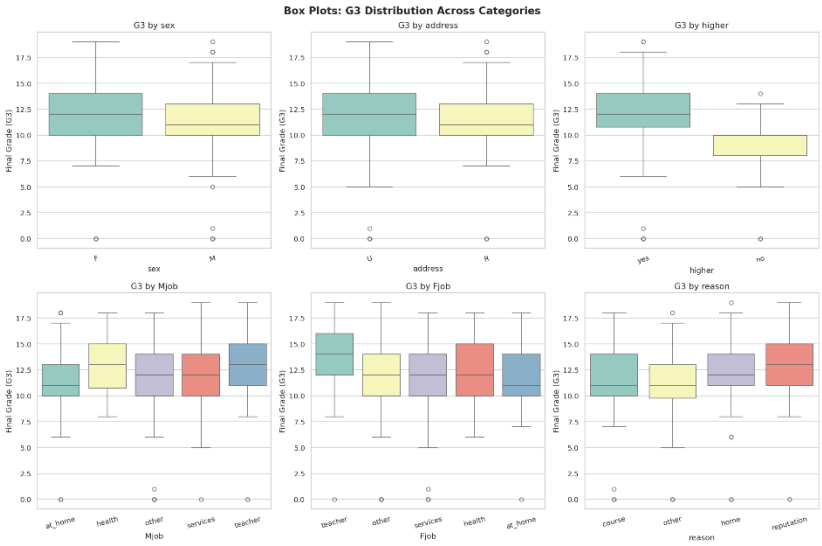

Fig 7. Box Plots: G3 Across Categories

Grade Progression (G1 → G2 → G3): 53.0% of students improved from G1 to G3, 28.5% stayed the same, and 18.5% declined. The mean grade change (G3 − G1) was +0.51.

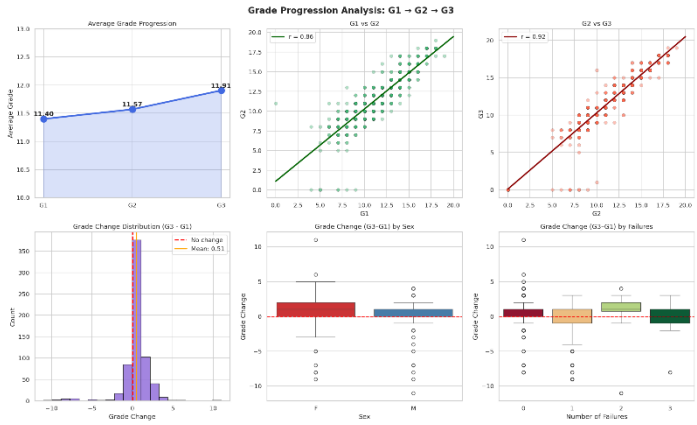

Fig 8. Grade Progression Analysis

Step 3: Feature Engineering

Five new features were created to capture composite student attributes:

Feature	Formula	Correlation with G3
avg_grade	(G1 + G2) / 2	0.905
grade_trend	G2 − G1	0.276
grade_change	G3 − G1	0.528
social_score	goout + freetime	−0.130
risk_score	failures + Dalc + Walc + absences/5	−0.294
support_score	sum of schoolsup, famsup, paid, internet	0.068
parent_edu	Medu + Fedu	~0.24
total_alc	Dalc + Walc	−0.209

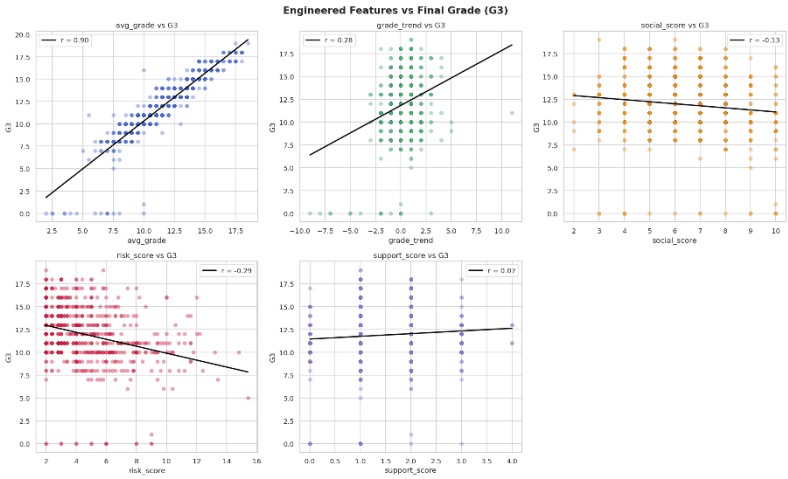

Fig 9. Engineered Feature vs Final Grade (G3)

Step 4: Data Preparation for Modelling

A binary target variable pass_fail was created (1 if G3 ≥ 10, else 0). The pass rate in the dataset was 84.6% (549 pass, 100 fail). All categorical columns were label-encoded. The dataset was split 80/20 into training (519 samples) and test (130 samples) sets using stratified splitting. Features were standardised using StandardScaler for Logistic Regression.

Step 5: Model Training and Evaluation

In this step, multiple machine learning models were trained and evaluated to predict student performance. The dataset was split into training and testing sets using a stratified 80/20 split, ensuring that the class distribution was maintained across both sets. This resulted in 519 samples for training and 130 samples for testing. To further ensure the robustness and generalization capability of the models, 5-fold Stratified Cross-Validation was also applied during training.

Three different models were implemented for comparison. Logistic Regression was trained using scaled features with a regularization parameter (C) of 1.0 and a maximum of 1000 iterations to ensure convergence. Random Forest was configured with 200 estimators, a maximum depth of 10, and a minimum of 2 samples per leaf, allowing it to capture complex patterns while avoiding overfitting. Gradient Boosting was implemented with 200 estimators, a learning rate of 0.05, and a maximum depth of 4, focusing on sequentially improving model performance.

The performance of each model was evaluated on the test dataset using multiple metrics, including Accuracy, Precision, Recall, F1-Score, and ROC-AUC. These metrics provided a comprehensive understanding of model effectiveness in terms of both overall correctness and class-wise prediction quality. Additionally, visual evaluation techniques such as confusion matrices, ROC curves, and Precision-Recall curves were generated to compare model performance and analyze classification behaviour in detail.

Results/Output

Model Performance on Test Set (130 samples):

Model	Accuracy	Precision	Recall	F1-Score	AUC
Logistic Regression	0.908	0.954	0.936	0.945	0.942
Random Forest	0.908	0.945	0.945	0.945	0.954
Gradient Boosting	0.908	0.938	0.955	0.946	0.955

All three models achieved identical test accuracy of 90.8%. Gradient Boosting achieved the highest AUC (0.955) and F1-Score (0.946), making it the best-performing model overall

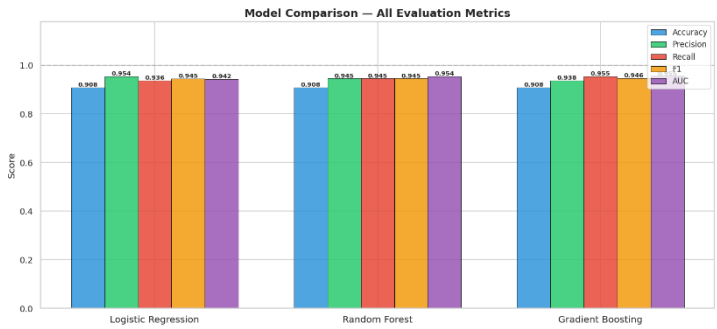

Fig 10. Model Comparison — All Evaluation Metrics

Cross-Validation Results (5-Fold Stratified):

Model	CV Accuracy	CV Std
Logistic Regression	0.9364	±0.0143
Random Forest	0.9268	±0.0199
Gradient Boosting	0.9210	±0.0295

Logistic Regression achieved the highest and most stable cross-validation accuracy (93.64% ± 1.43%), suggesting it generalizes most consistently across folds despite identical test set performance.

Confusion Matrices:

All three models correctly classified the majority of students. On the 130-sample test set (20 actual failures, 110 actual passes):

Model	True Negatives (Fail correctly predicted)	False Positives	False Negatives	True Positives (Pass correctly predicted)
Logistic Regression	15	5	7	103
Random Forest	14	6	6	104
Gradient Boosting	13	7	5	105

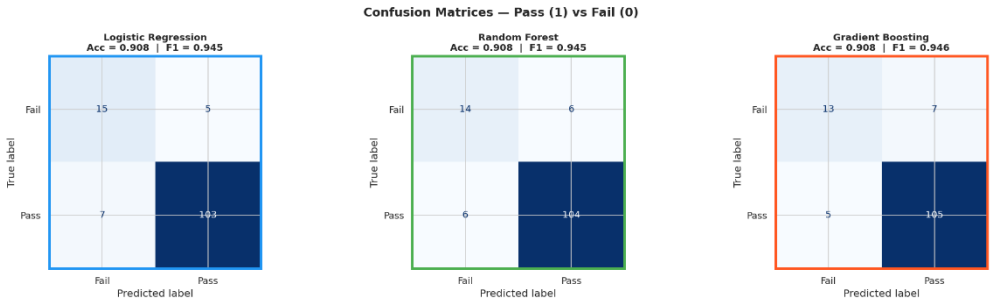

Fig 11. Confusion Matrices — Pass (1) vs Fail (0)

ROC-AUC Curves:

All models significantly outperformed the random baseline (AUC = 0.500). Gradient Boosting (AUC = 0.955) and Random Forest (AUC = 0.954) were nearly identical and slightly better than Logistic Regression (AUC = 0.942).

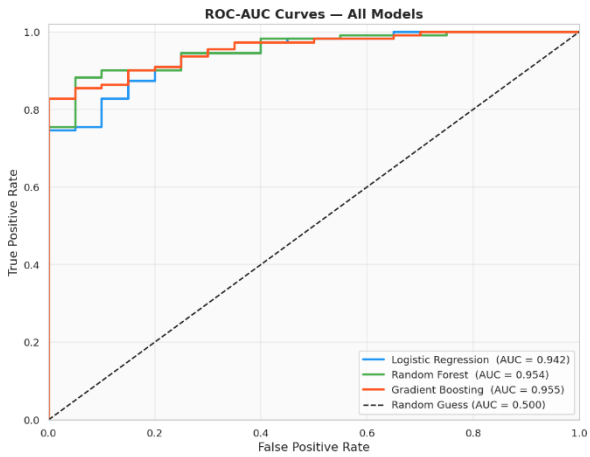

Fig 12. ROC-AUC Curves — All Models

Precision-Recall Curves:

With the no-skill baseline at 0.846 (reflecting the dataset's 84.6% pass rate), all models achieved very high Average Precision scores: Logistic Regression (AP = 0.989), Random Forest (AP = 0.992), and Gradient Boosting (AP = 0.992).

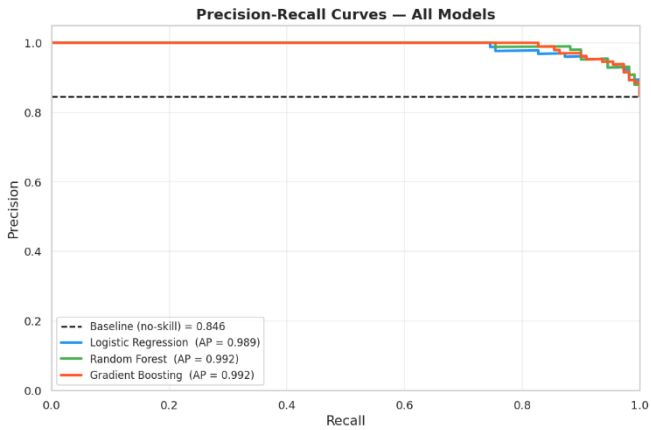

Fig 13. Precision-Recall Curves — All Models

EDA Summary — Top Correlators with G3 (keep this table from before):

Feature	Correlation with G3
avg_grade	+0.905
grade_change	+0.528
higher (aspiration)	+0.332
grade_trend	+0.276
studytime	+0.250
failures	−0.393
risk_score	−0.294
school (GP vs MS)	−0.284
total_alc	−0.209

Dalc	−0.205

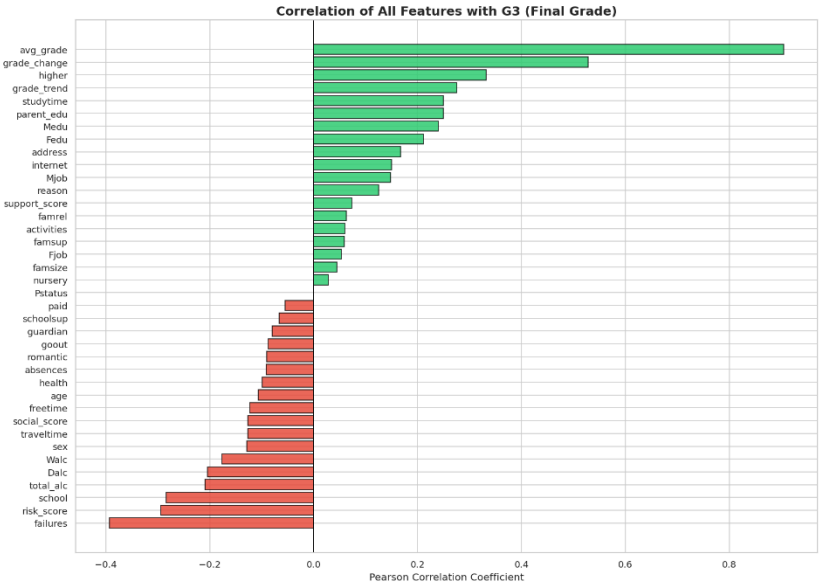

Fig 14. Correlation of All Features with G3

Grade Progression Stats

- Avg G1: 11.40 | Avg G2: 11.57 | Avg G3: 11.91
- Students who improved (G3 > G1): 344 (53.0%)
- Students who declined (G3 < G1): 120 (18.5%)
- Students who stayed same: 185 (28.5%)

Conclusion

This project successfully combined extensive exploratory data analysis with machine learning classification to understand and predict student academic performance. The EDA revealed that prior grades (G1, G2) are the strongest predictors of G3, while past failures (r = −0.39) and alcohol consumption are the most significant negative factors. All three classifiers — Logistic Regression, Random Forest, and Gradient Boosting — achieved a test accuracy of 90.8% and F1-scores above 0.945, well above the no-skill baseline of 84.6%. Gradient Boosting slightly outperformed on AUC (0.955) while Logistic Regression showed the best and most stable cross-validation accuracy (93.64%). The results confirm that student pass/fail outcomes can be predicted reliably using a combination of academic history, lifestyle features, and engineered risk indicators.

Future Scope

In future work, deep learning models such as neural networks could be explored for better accuracy, and the binary classifier could be extended into a multi-class grade predictor for more granular insights. The system could also be deployed as a web application where teachers input student data at the start of each term to receive instant at-risk alerts, enabling early intervention before grades decline.

References

[1] Student Performance Dataset – Kaggle, https://www.kaggle.com/datasets/devansodariya/student-performance-data

[2] Scikit-learn: Machine Learning in Python — Official Documentation, https://scikit-learn.org/stable/